

## Week-4

- Demand Response Curve
- Simple Linear Regression

Good → Price

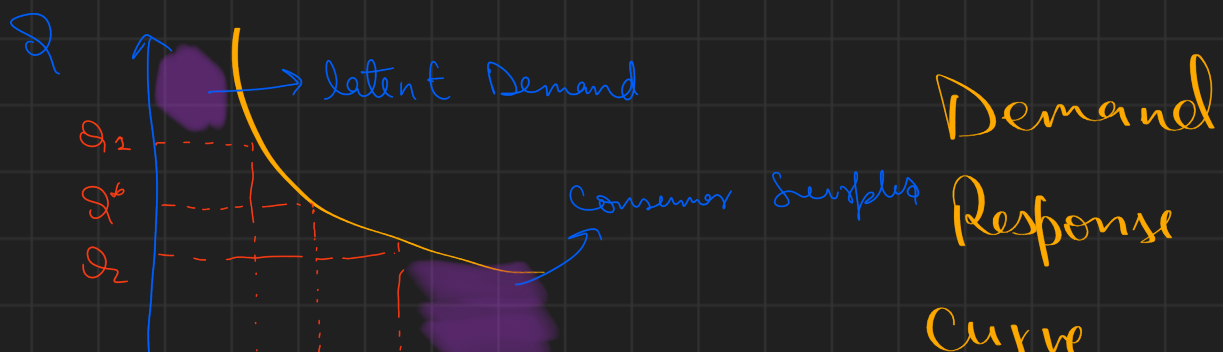
Demand → Number of Goods we purchase

$P \uparrow \quad D \downarrow$

- Normal goods:  $I \uparrow \quad D \uparrow$
- Inferior goods:  $I \uparrow \quad D \downarrow$

## Law of Demand:

For Normal goods: Price of good is inversely  $\propto$  to  $Q_d$  demanded.



## Properties of DR curve:

- Continuous
- Differentiable
- Non-negative
- Downward Slope

Latent Demand: Hidden Demand / Demand Not Satisfied  
(Hidden)

Increases w/ Price

Consumer Surplus: Extra Utility enjoyed by consumer because price < what they can afford (WTP)

Elasticity:  $\frac{\% \text{ change in demand}}{\% \text{ change in Price}} = \frac{\frac{\Delta D}{D} \times 100}{\frac{\Delta P}{P} \times 100}$

$$E = \frac{\frac{\Delta Q}{Q} \cdot \frac{P}{\Delta P}}{\frac{\Delta Q}{\Delta P} \cdot \frac{P}{Q}} = \frac{Q_f - Q_i}{P_f - P_i} \times \frac{P_i}{Q_i}$$

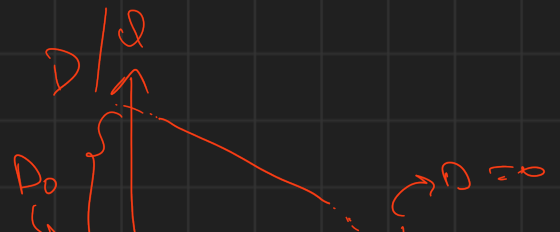
$$\lim_{\Delta P \rightarrow 0} E = \frac{\frac{dQ}{dP} \cdot \frac{P}{Q}}{\frac{dQ}{dP} \cdot \frac{P}{Q}} =$$

$\left(\frac{\Delta Q}{Q}\right) / \left(\frac{\Delta P}{P}\right) \rightarrow$  Elasticity b/w 2 pts

Linear Demand Response Curve:

$$D(p) = D_0 - m p$$

$\hookrightarrow$  as  $p \rightarrow 0 \rightarrow D = m$



$$\epsilon = \left. \frac{dQ}{dP} \right|_{P=P} \cdot \frac{P}{Q} = \frac{-mP}{D_0 - mP}$$

$$\epsilon > 1$$

Elastic

$$\frac{\Delta Q}{Q} > \frac{\Delta P}{P}$$

$$\epsilon = 1$$

Unit elastic

$$\frac{\Delta Q}{Q} = \frac{\Delta P}{P}$$

$$\epsilon < 1$$

Inelastic

$$\frac{\Delta Q}{Q} < \frac{\Delta P}{P}$$

$\epsilon < 1 \rightarrow$  Simply  $P \uparrow$  to inc. Revenue

$\epsilon > 1 \rightarrow$  Revenue increases by Deer price  
(closer 0)

$$(R = D(p) \times p)$$

Linear Demand Response Curve:

$$D(p) = D_0 - mP$$

$$D_0 = D(p=0) \text{ \{ Market Size \}}$$

$$p_s \rightarrow D(p=p_s) = 0 \text{ or } \Rightarrow p_s = D_0/m$$

$$\epsilon = \text{Magnitude of } \left( \frac{\Delta Q}{Q} / \frac{\Delta P}{P} \right) \text{ or } \frac{mP}{D_0 - mP}$$

In Linear D-R curve:  $\epsilon \neq$  constant

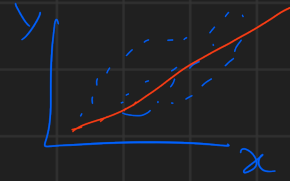
$$\epsilon(p=0) = 0$$

$$\epsilon(p=p_s) = \frac{mP}{D_0 - mP} \text{ as } P \rightarrow D_0/m \rightarrow \infty$$

# SLR $\rightarrow$ Simple Linear Regression

I/p var  $\rightarrow x$

O/p var  $\rightarrow y$

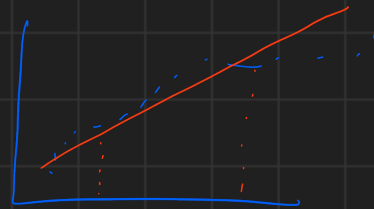


$$y = \beta_0 + \beta_1 x_1$$

| $y$        | $x$         |
|------------|-------------|
| Dependent  | Independent |
| Explained  | Explanatory |
| Predicted  | Predictor   |
| Regressand | Regressor   |

$$y = \beta_0 + \beta_1 x$$

$$y = mx + c$$

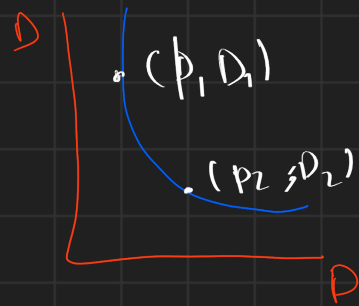


$$y = \beta_0 + \beta_1 x + \epsilon_i \quad ; \quad \epsilon \sim \mathcal{N}(0, \sigma^2)$$

$$\textcircled{1} \quad D(p) = D_0 - mp$$

| $p$ | $D$ | $D$ |
|-----|-----|-----|
| 1   | 10  |     |
| 2   | 8   |     |
| 3   | 7   |     |
| 4   | 5   |     |
| 5   | 3   |     |

## Non Linear Demand Response Curve:



→ Elasticity is constant

→  $D(p) = C p^{-\epsilon}$  ↙ elasticity

$$D_1 = C p_1^{-\epsilon}$$

$$D_2 = C p_2^{-\epsilon}$$

$$D_1/D_2 = (p_1/p_2)^{-\epsilon} \Rightarrow$$

$$\log(D_1/D_2) = -\epsilon \log(p_1/p_2)$$

$$\epsilon = -\log(D_1/D_2) / \log(p_1/p_2)$$

$$\epsilon = +\log(D_1/D_2) / \log(p_2/p_1)$$

Historical Records:

| P   | D   |
|-----|-----|
| ... | ... |
| ... | ... |
| ... | ... |
| ... | ... |

Apply Log-Log Transform

$$D(p) = C \cdot p^{-\epsilon} \Rightarrow \ln(D) = \boxed{\ln C} - \boxed{\epsilon} \ln p$$

Intercept =  $\ln C$  or  $C = e^{\text{Intercept}}$

Slope =  $\beta_1 = -\epsilon \rightarrow \text{elasticity} = -\text{slope}$

$\epsilon \rightarrow$

$C \rightarrow$

P vs Q<sub>d</sub>

1

The linear demand response for product-A is modeled as a simple linear regression represented as  $D(P) = 1500 - 20 \cdot P$ , where  $D(P)$  is the demand at price- $P$ . Then, answer the following questions:

(a)  
(b)  
(c)

- What is the elasticity of the demand response curve when the price is Rs.50?  
What is the satiating price for the demand response curve?  
What is the Market Size for the demand response curve?

$$D = D_0 - mp \quad (D_0 = 1500; m = 20)$$

$$(a) \quad e(p) = \frac{mp}{D_0 - mp} = \frac{20 \times 50}{1500 - 20 \times 50} = \frac{1000}{500} = 2$$

$$(b) \quad 0 = 1500 - 20 \times p \Rightarrow P_s = 1500/20 = 75 \text{ Rs.}$$

$$(c) \quad D_0 = \text{Market Size} = 1500$$

2

Your local grosser has observed that the demand for ButterMilk follows a linear demand response curve. Hence, he has decided to raise the "Butter Milk" price by Rs. 5 rupees per bottle to Rs. 15 per bottle. If the demand was 100 units per day before the rise and is now 80 units per day after the rise, then answer the following questions

- a) What is the elasticity of demand?  
b) Given the computed elasticity, what can you say about the "Butter Milk" demand? (choose all that are applicable)

It is elastic

It is inelastic

It is luxury

It is a necessity

$$(a) \Rightarrow e = \left| \frac{\Delta D / D_i}{\Delta P / P_i} \right| = 0.4$$

$$(b) \quad e < 1 \rightarrow \text{Inelastic}$$

3

For a demand response curve which has constant elasticity, which of the following statements are true (choose all that are applicable)

If the curve is for an inelastic product, the revenue is increased only by setting price close to zero

If the curve is for an inelastic product, the revenue is increased by simply increasing the prices

If the curve is for an elastic product, the revenue is increased by simply increasing the prices

$$\frac{\Delta Q}{Q}$$

$$\frac{\Delta P}{P}$$

3

For a linear demand response curve, the satiating price is \_\_\_\_

The price beyond which latent demand exists

The price at which demand is zero

The price beyond which the consumer surplus exists

3

Select all the null hypotheses for a Simple Linear Regression model.

- ☐ The intercept is not significant ( $H_0 : \beta_0 = 0$ )
- ☐ The slope is not significant ( $H_0 : \beta_1 = 0$ )
- ☐ The intercept is significant ( $H_0 : \beta_0 \neq 0$ )
- ☐ The slope is significant ( $H_0 : \beta_1 \neq 0$ )

$$H_0 : \beta_0 \neq 0$$

$$H_0 : \beta_1 \neq 0$$

IITM-BA's courses elasticity for number of student registrations is expected to be constant at 0.9 for the next couple of years. Currently the course fee is Rs.10,000 and 1500 students have registered. If the course fee is changes to Rs. 8000, how many students will register?

1200

1770

1230

$$0.9 = \left| \frac{\Delta Q/Q}{\Delta P/P} \right| \Rightarrow 0.9 = \left| \frac{\Delta Q/Q}{-0.2} \right| \Rightarrow 0.9 = \left| \frac{Q_f - 1500}{1500} \right| \frac{1}{-0.2}$$

$$0.9 \times 0.2 = \frac{Q_f - 1500}{1500} \Rightarrow Q_f = 1500 + 270 = 1770$$

1

What is the satiating price for a Constant-elasticity price response function?

Zero

Does not exist

Equal to the Market size (D0)

8

Using the excel regression output given in Figure-2 answer the given sub questions.

| SUMMARY OUTPUT        |          |            |    |    |
|-----------------------|----------|------------|----|----|
| Regression Statistics |          |            |    |    |
| Multiple R            | 0.966016 |            |    |    |
| R Square              | 0.933186 |            |    |    |
| Adjusted R Square     | 0.926505 |            |    |    |
| Standard Error        | 363.8778 |            |    |    |
| Observations          | X1       |            |    |    |
| ANOVA                 |          |            |    |    |
|                       | df       | SS         | MS | F  |
| Regression            | 1        | X2         |    | X4 |
| Residual              | 10       | 1324070.51 | X3 |    |
| Total                 |          | 19817291.7 |    |    |

Figure-2

$$X1: n - 1 \text{ of } 06 = n$$

$$\& \text{ Total } df = n - 1$$

$$\Rightarrow X1 = 11 + 1$$

$$SS_{reg} + SS_{res} = SS_{total}$$

$$\chi^2 = 19817291.7 - 1324070.51$$

$$\chi^2 = 18493221.19$$

$$s_y = MS_{Res} = \frac{SS_{Res}}{df_{Res}} = 132407.051$$

$$dy \Rightarrow F_{stat} = \frac{MS_{reg}}{MS_{Res}} = 139.669$$

9

If the demand response curve for the data given in the following table is modeled by a constant elasticity curve, then answer the following questions

| Demand (number of units) | Price (Rs.)         |
|--------------------------|---------------------|
| 0.50                     | 20 $\rightarrow +1$ |
| 0.33                     | 30                  |
| 0.25                     | 40 $\rightarrow +2$ |
| 0.20                     | 50                  |

$$D(p) = C \cdot p^{-\epsilon} \quad \text{Elasticity}$$

- a) What is the demand when the price is Rs.1?  
b) What is the price elasticity for the demand response curve?

$$(P_1, D_1) = (20, 0.5) ; (P_2, D_2) = (40, 0.25)$$

$$D_2 = C p_2^{-\epsilon} \quad \& \quad D_1 = C p_1^{-\epsilon}$$

$$\epsilon = \frac{\ln(D_1/D_2)}{\ln(P_2/P_1)} = \frac{0.69314}{0.69314} = 1$$

$$C=1 \quad D = C \cdot p^{-1} \quad \text{or} \quad D = \frac{C}{p}$$